1. Clinical Question & Core Hypothesis

Clinical Question

Why do tumors with immune infiltration sometimes fail to confer a survival benefit? The provided path suggests a link between loss of epithelial adhesion (E-cadherin/CDH1) and immune-checkpoint biology (LAG3), implying that epithelial state may reshape T cell function. 2) Tier 1 paths used: (a) CDH1 down-regulation → EMT/invasion (mechanistic backbone that can alter tumor–immune interfaces); (b) BLCA↔LAG3 edge interpreted as representation of tumor-type–specific modulation of LAG3 and, more generally, that tumor biology can control LAG3 expression/function. I built on these because they provide the most direct mechanistic route linking epithelial integrity to checkpoint-mediated T cell dysfunction. 3) Tier 2 evidence supporting the claim: tissue-specific variation in LAG3 (e.g., low in READ) argues checkpoint expression is context-dependent; known T-cell exhaustion genes (PDCD1, CTLA4, TIGIT) and stromal/matricellular genes (COL1A1, FN1) can mediate immune exclusion or chronic activation—these reinforce how loss of epithelial architecture could produce a pro-exhaustion microenvironment. 4) Choice of endpoint: Overall survival (OS) was selected because TCGA-BRCA contains robust OS data and because the question asks why immune infiltration sometimes fails to translate into improved long-term outcomes. 5) Deliberate exclusions: I excluded organ- and method-specific nodes (bladder, ultrasound) as they are not applicable to TCGA-BRCA; I also excluded broader correlative gene lists from the core hypothesis to maintain focus (they are listed as secondary_modifiers)?

Working Hypothesis

In breast cancer, loss of epithelial integrity driven by CDH1 downregulation promotes EMT and remodeling of the tumor microenvironment (including ECM/stromal activation with COL1A1 and FN1), which increases chronic antigen exposure and T cell stimulation. This environment induces an exhaustion-like program in tumor-infiltrating CD8+ T cells, marked by elevated LAG3 expression. Consequently, tumors with low CDH1 and high LAG3 harbor an infiltrate that is numerically present but functionally impaired, correlating with poorer overall survival despite robust TIL density. The strength and detectability of this axis vary by molecular subtype and spatial context, with the highest LAG3 enrichment in CDH1-low, ECM-rich tumor regions. If surrogate blockade of LAG3 or combination immunotherapies are applied, these CDH1-low/ECM-rich tumors may show restored T cell function, suggesting a context-dependent, targetable immune dysfunction linked to epithelial state.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CDH1, LAG3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CDH1: CDH1 down-regulation → EMT/invasion (mechanistic backbone that can alter tumor–immune interfaces); (b) BLCA↔LAG3 edge interpreted as representation of tumor-type–specific modulation of LAG3 and, more generally, that tumor biology can control LAG3 expression/function.

LAG3: LAG3 edge interpreted as representation of tumor-type–specific modulation of LAG3 and, more generally, that tumor biology can control LAG3 expression/function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	TIGIT	Supporting modifier from knowledge graph
Secondary	FN1	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: that shape the immune microenvironment and can promote immune exclusion or chronic stimulation. LAG3 is an inhibitory receptor associated with T cell exhaustion and can be upregulated in chronic antigen exposure; its expression is context-dependent across breast cancer subtypes.

Prediction 2: . This yields a testable model: CDH1 low status predicts higher LAG3 on CD8+ T cells in ECM-rich regions, correlating with worse OS.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PDCD1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTLA4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.